

An Evidence-Driven ePolyScat Research Portal

End-to-End Scientific Workflows, Reproducible Inputs, and Verified Computation

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PROJECT ACCEPTANCE
REVIEW
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Scientific Gateway Delivery

RESEARCH GAP

Scientific workflows span specialized programs, remote schedulers, and local analysis tools. Job completion alone does not prove that scientifically valid outputs were produced or that the correct files reached the next stage.

ACCEPTANCE CLAIM

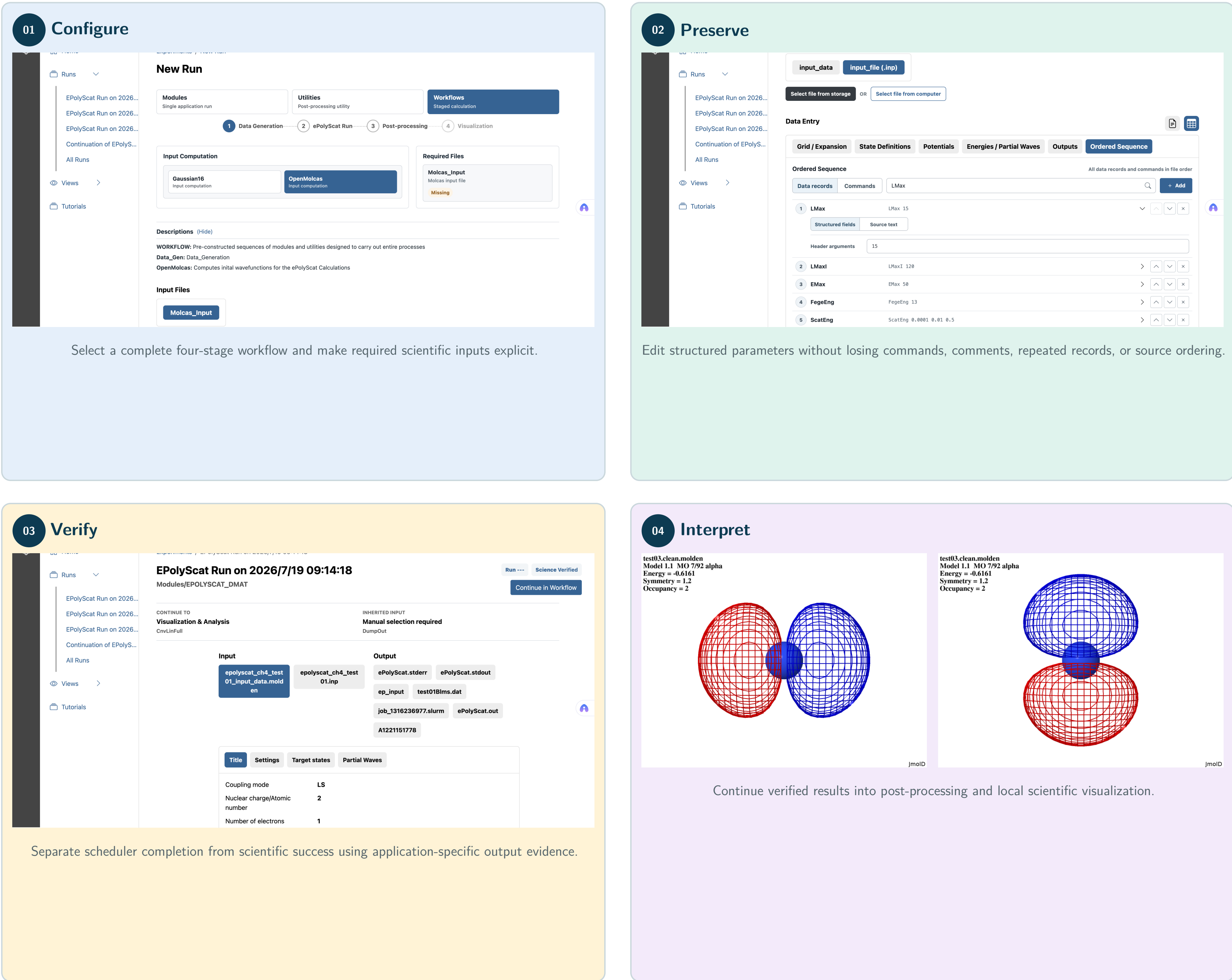
Connect configuration, execution, scientific validation, and stage-to-stage handoff in one portal experience.

Workflow Portal and Scientific Verification

KEY CONTRIBUTION

We transformed ePolyScat from a collection of disconnected tools into a guided, inspectable workflow. Each stage preserves the researcher's original input, records output provenance, and exposes a verified handoff to the next calculation or visualization step.

Evidence narrative: from input to interpretation



Engineering decisions

- Lossless ordered document model for reproducible ePolyScat inputs.
- Semantic output roles instead of filename-only inference.
- Resource-aware execution contracts across supported HPC systems.
- Provenance-aware continuation with researcher review before submission.

Scientific value

Researchers can reproduce exactly what was submitted, distinguish scheduler artifacts from scientific results, and continue a calculation without manually rebuilding file dependencies.

Reproducible input → interpretable output

ACCEPTANCE RESULT

Portal configuration → HPC execution → scientific proof → provenance-aware handoff → local visualization is operational and demonstrated with real runs.

Acceptance evidence



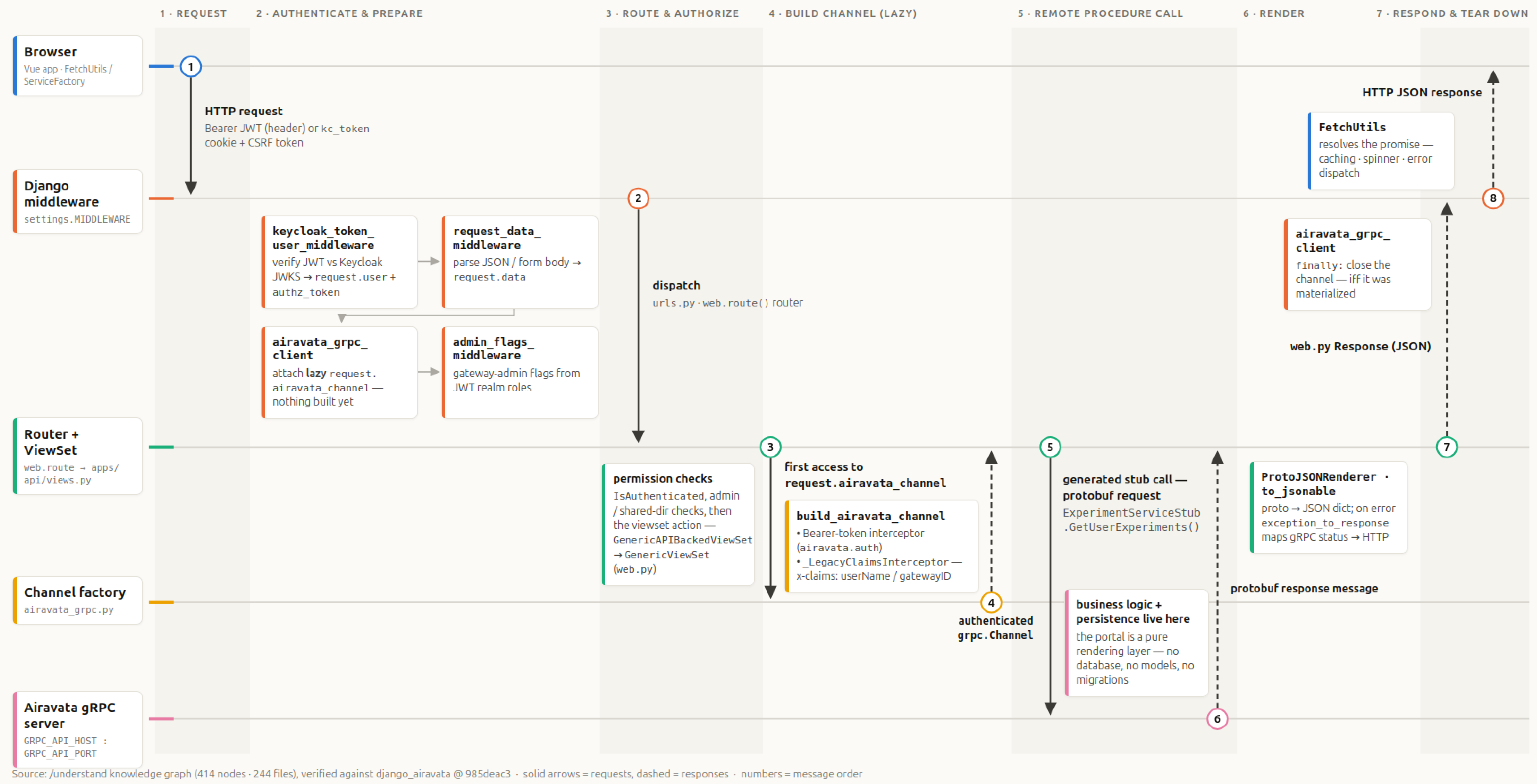
Non-gRPC test coverage, HPC wrapper regressions, and the production build passed successfully.

Runs Management Service and New Server (gRPC) Migration

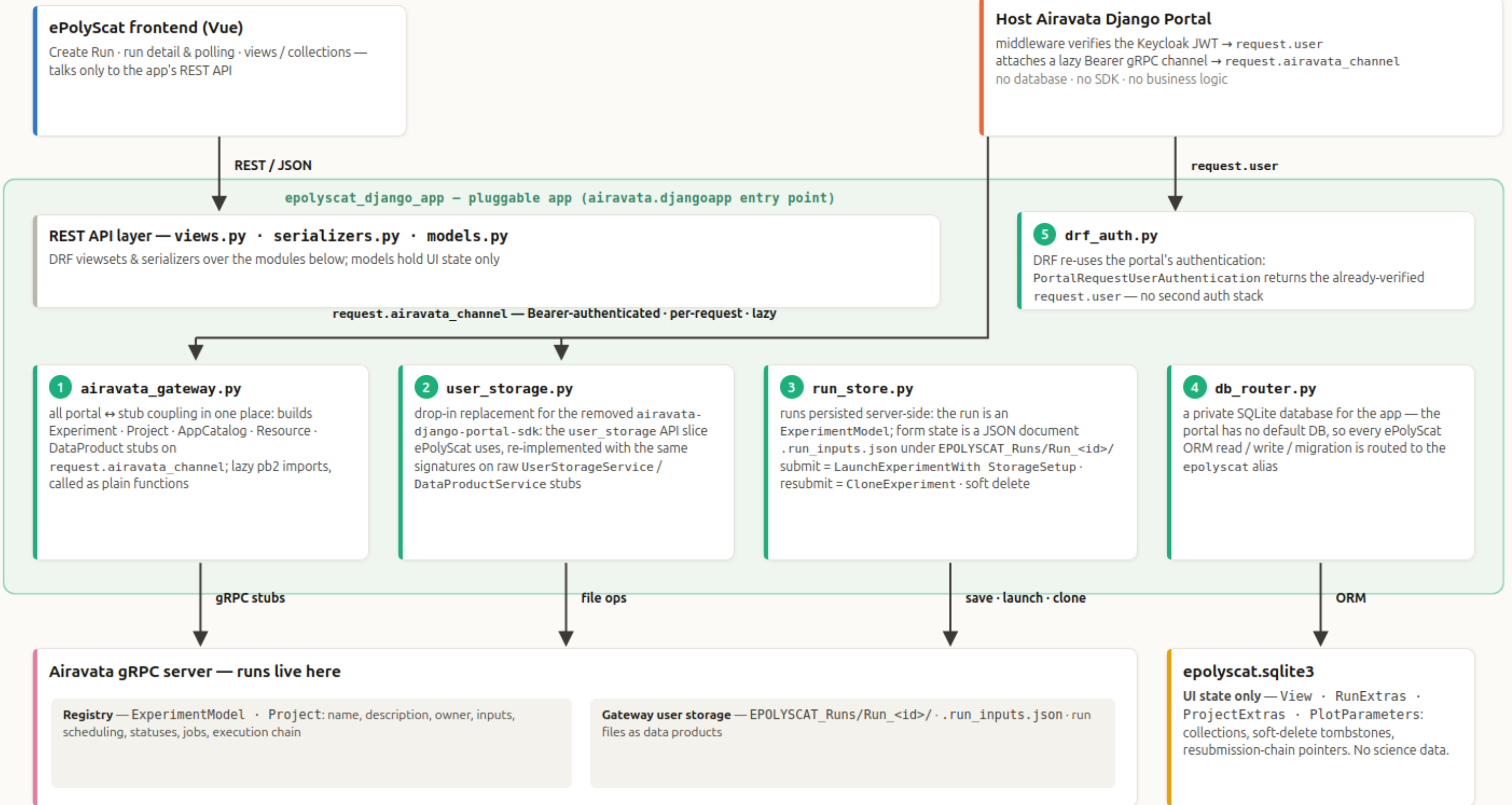
KEY CONTRIBUTION

- Users can view all runs under All Runs—each showing its name, status, resource, and clone or delete actions—and group selected runs into a View from the Runs page. Within a View, users can select, delete, or open its runs. Users can also create an Experiment, add new runs inside it, and delete experiments they no longer need.
- Refactor (network): Migrate application from Thrift server to gRPC server.

gRPC Request/Response Lifecycle



Engineering Design



Benefits of adapting to gRPC server

- Easier to maintain source code: When using Thrift, every request calls the SDK. In contrast, when using gRPC, since the SDK is deleted, requests call the server directly. In addition, gRPC codebase is significantly smaller than Thrift codebase. As a result, source code is less complicated and easier to maintain and debug.
- Higher consistency between client and server: The .proto files are now the single source of truth for both the server and the client. This helps the client and server avoid drifting apart. Protobuf is a widely-used industry standard, while Thrift is developed by the Airavata team only.
- Easier to add plugins: gRPC can include hooks for every request, for example attaching the Keycloak login token onto each request for authorization, by using just one file instead of injecting it manually everywhere.

Broader Impacts and Future Outlook

Together the two contributions turn ePolyScat into a gateway pattern other projects can reuse: a guided workflow that preserves researcher input and proves scientific validity, running over a thin portal layer that talks to Airavata through a shared gRPC contract. Because inputs stay lossless and outputs carry provenance, results remain reproducible and reviewable; because the files are the single source of truth, new services can be added without renegotiating the client-server boundary.

Next steps:

- Broaden verified-output coverage to more ePolyScat calculation types and HPC resources.
- Extend run management with richer filtering, sharing, and experiment-level provenance in the UI.
- Move the gRPC server into production.
- Develop other features for the Airavata portal, for example interactive notebooks and AI-assisted chatbot.